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AMAZON ROBOTICS LLC

300 RIVERPARK DRIVE

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Rev.	Description of Change	Sections Affected	Originator/Department
01	Initial release	ALL	P. Murphy / TechPubs
02	1. Add "flex drop" to Pathway Infrastructure 2. Double-duplex convenience outlets replaced with "at least one" outlet receptacle. 3. Label external receptacles and disconnect switches on charger power distribution box. 4. Clarify data connections to MSC for large systems 5. Remove heat displacement from deprecated part in MSC. 6. Added MSC dimension print for reference 7. Add FA connections for reference 8. Rebrand Amazon Robotics	5.1, 6.1, 7.1, 8, 11, 13.2, 13.9	G. Tierney / Engineering
03	1. Updated "Battery Charging Stations" section to lower current requirements and allow 50/60Hz input. 2. Replaced all references to "Kiva" with references to Amazon Robotics (AR).	8	C. Ropi / Engineering
04	Updated FA drawing with NO contacts to be consistent with description	13.9	M. Chung / Engineering
05	Various updates to sections to align with latest approved revision of the NA Electrical specification; specific section updates below. 1. Removed KCS reference 2. Removed labeling specifics for data and power. 3. Added location / height dimension for workstation and PSC related receptacles. 4. Updated part 1 of this section to reference table (13.3) for electrical convenience outlets. 5. Removed PSC location section 6. Updated section to add requirements for FCX. 7. Removed Data connection specifics and referenced the network plan 8. Updated to align with NA spec 9. Removed references to KCS 10. Removed Network Diagram 11. Removed PSC Dimensions 12. Removed Battery Charging Station Power Distribution Box Location	ALL 1.0 5.4 5.5 6.1 7.3 9.1 9.2, 11.2 9.3, 9.4 9.0, 10.0, 11.0 13.1 13.3 13.5	B. Klein / Engineering P. Freedman / Prod IT

	13. Removed R Series Charging Station Dimensions	13.6	
	14. Removed G Series Charging Station Dimensions	13.7	
06	Added 1737-SUPP-VBIV5 – <i>Module drawing for VBI V5 Tower</i> Added VBI specific acronyms to this section. Removed DC UPS acronym as this is not used anymore. Updated power requirements based on the Nike 2.0 station current draw at 100VAC. Added verbiage about ganging outlets and to comply with local / regional electric codes. Also, added requirement for double-duplex JIS 8303 receptacles for convenience use. Updated frequency of power to be 50 or 60 Hz since both are used in Japan. Modified circuit breaker from 15 AMP to 20 AMP because the L5-20R is a 20A receptacle. Removed reference to DC UPS circuits and replaced with PSC circuits. Deleted thermal output per PIT's request Deleted section 10 for the provisioning server. This does not drive requirements to the building infrastructure as the provisioning server plugs into the FCX PDU. Updated frequency of power to be 50 or 60 Hz since both are used in Japan. Updated the voltage from 120VAC to 100VAC to comply with Japanese standards Added new section 12.0 for VBI Electrical Requirements	2.0 4.0 6.1 7.1 8.1 9.3 10.0 11.1 12.0	B. Klein / Engineering P. Freedman / PIT
07	Updated section to call out VBI as another item that shall not share power distribution with the PSC. Updated section to allow ganging of PSC's as long as the current draw and local / regional electric codes are considered. Removed "Gate" from the section header to align with NA and EU specs.	7.0	B. Klein / Engineering
08	Changed revision from 07 to 08 in the header. Updated revision, date, and copyright date. Added all building types that this specification will apply to which includes Sortable Fulfillment centers, Pantry Fulfillment Centers and Sort Centers.	ALL Cover 1.0	B. Klein / Engineering M. Chung / Engineering

	Added AR Workstation and PSC power drop heights for all building types includes Sortable Fulfillment centers, Pantry Fulfillment Centers and Sort Centers.	5.5	
	Added language to allow drops for AR Workstations and PSC's to be removed and relocated without the need for a qualified electrician.	5.5	
	Added reference for VBI data and power drop heights.	5.5	
	Added units of feet to align with wording in other electrical documents.	9.4	
	Updated temperature range to align with FCX requirements	10.3	
	Changed power cord length to 3048mm and updated the location of the AC inlet for the MSC	10.4	
	Updated the language in the table to remove reference to the MDF room and to clarify the operation of the fire alarm and interconnect requirements	11	
	Changed VBI power requirements to specify a single receptacle	12.2	
	Updated VBI data and power drop heights to 2133.6 mm (84") based on request from Joe Matthew.	12.4	
	Updated Power Distribution Box picture.	13.2	
	Updated the title of the drawing to clarify the normal state of operation for the fire alarm and connections to the MSC	13.4	
09	Changed revision from 08 to 096 in the header.	ALL	B. Klein / Engineering
	Added table of figures and linked to figures within the document.	ALL	
	Updated table of contents	ALL	
	Updated revision, date, and copyright date.	Cover	
	Updated the AR Workstation Types – Changed Sort Center to Mobile Sortation Solution (Elevated and Ground). Added Pallet Sortation and Air Gateway. Also, added Robin (Robotic Induct).	1.0	
	Updated section 2.0 to include a map of Japans power grids showing Japan East (50Hz) versus Japan West (60Hz).	2.0	
	Updated section 3.0 to add verbiage indicating that local / national codes may supersede requirements in the document.	3.0	

	Update section with new requirements for hanging power drops / flex drops for the NA.	5.0	
	Separated general requirements for power drops and data drops. Updated entire section to align with recent updates to the EU electrical spec.	5.0	
	Updated figure references to match to the new table of figures.	5.1	
	Added AR Workstation and PSC power drop heights for all building types includes Sortable Fulfillment centers, Pantry Fulfillment Centers, Mobile Sortation Solution (Elevated and Ground), Pallet Sortation and Air Gateway.	5.4	
	Updated header text to reference the network plan for number of FCX cabinets. Also, updated table heading from KCS to FCX. Also, updated table formatting to align with the rest of the document.	9.0	
	Update sections to reference 50 or 60Hz, clarified voltage and current where needed for 100VAC circuits.	8.1, 8.3	
	Added new section 13 to include power and data drop requirements for the Robotic Induct Station (Robin)	13.0	
	Appendix section changed from 13 to 14.	14.0	
	Added new section 14.5 which includes example picture of data drops in the ceiling. Request from SIM ticket ARHWStations-2009.	14.5	
	Added new section 14.6 with example of Flex drops	14.6	
10	Added new buildings to include Middle Mile Sort Center and AMZL. https://issues.amazon.com/issues/ARHWStations-2460	1.0, 5.4	B. Klein / Engineering 06/26/20
	Update overview to be more generic and cover all AR enabled buildings.	1.0	
	Added KIX5 White paper reference that describes overvoltage issue root cause & corrective action.	2.0, 8.0	
	Added AMZL and AR to the list of acronyms.	4.0	
	Added language "It is permissible to move the receptacle to accommodate better cable management". This is to enable drop height changes for the ARSAW stations in sortable FC's. Request originated from the AR Deployment team. This will also apply to sort center stations.	5.4	

	Added recommended implementation language for power receptacles; this is intended to avoid confusion based on request from Deployment.	6.1	
	Updated section to add spare data drop ¹ and clarification on language as request by COMET and Hargis.	6.2	
	Updated section and added clarifying language by COMET and Hargis.	7.2	
	Provided provisions to use unistrut instead of a column in the event a column is not available for mounting the power distribution box.	8.2, 8.4	
	Added language "The distribution box shall be located on the same side of the fence where the charger is located." https://issues.amazon.com/V155763117 . Changed KIVA to AR.	8.4	
	Added clarifying language for origination of the UPS circuit based on input from Deployment.	10.1	
	Update title wording for clarification	13.1, 13.2	
	For the Robin station, clarified language for data drop and added a requirement for a second data drop for connection to the safety network.	13.5	
	Added clarifying language for Robin data drop location.	13.6	
11	Added clarification that SSD sites are part of sortable fulfillment centers.	1.0	Bill Klein / Engineering 07/28/21
	Added new acronym for Sub-same day (SSD) sites.	4.0	
	Update section to correct figure references.	5.1	
	Added clarification to allow movement of the drop both horizontal or vertical to allow better cable management.	5.4a	
	Updated section to clarify that the power drop may be moved, vertical or horizontal, if required. This was added for clarification purposes.	5.6	

¹ IT use case for spare data port – Monitoring/Management port for data facilities devices (UPS or PDU) running the Nike cabinet. Availability of the data facilities management VLAN requires these devices be directly run back to an IDF. This would allow for lights out support and remote monitoring for better preventative maintenance/incident response of the cabinet UPS.

	Updated in accordance with https://issues.amazon.com/issues/HW-SUS-180 to add flexibility for data drop mounting heights for high-bay (>3658mm ceiling) building types.	8.0	
	Added clarification on branch circuit method versus power distribution method.	8.0	
	Updated entire section to clarify new electrical power requirements and methods of mounting them for the Robin station.	13.1	
12	Removed 'metal bracket'	5.1.3	M. DellaCroce / Engineering 08/01/22
	Updated maximum box size to 150mm x 150mm x 104mm	5.3.1	
	Updated chargers receptacle and power distribution height location to range of 1200mm to 1900mm	5.4.4	
	Added "unless otherwise specified" to data drops at ceiling location.	5.6.2	
	VBI receptacle location updated to 2130mm above the service level and data joist/ceiling mounted is clarified as lowest ceiling truss such that the data port is not located on the underside of the corrugated decking.	12.4	
	Robin power and data mounting locations updated to 820mm minimum above service level, hard requirement regarding power disconnect removed in favor of language permitting installation in accordance with local / regional electrical code.	13	

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1. Overview

The purpose of this document is to provide power and data wiring requirements for an Amazon Robotics enabled buildings in Japan. Amazon Robotics will not provide or install any of the materials indicated by this document unless otherwise noted. The requirements pertain to the following:

- AR Workstations
 - Sortable Fulfillment Centers (Includes Sub-Same Day SSD buildings)
 - Pantry Fulfillment Centers
 - Mobile Sortation Solution – Ground Floor
 - Mobile Sortation Solution – Elevated Floor
 - Pallet Sortation – S-Drive
 - Air Gateway
 - Middle Mile Sortation Center
 - Amazon Logistics (AMZL) Delivery Station
- PSC (Perimeter Safety Controller)
- Battery Charging Stations
- FCX (FC Execution System)
- Provisioning Server
- MSC (Main Safety Controller)
- VBI (Visual Bin Inspection)
- Robin (Robotic Induct)
- Fire Alarms

2. Reference Documents & Images

- 810-00406: *Floor Safety System HW Manual*
- 965-0039-03: *Charger Installation Instructions*
- 610-00629: *Assembly Instructions Main Safety Controller*
- 700-0101-002: *Assembly Instructions Gate Safety Equipment*
- 1737-SUPP-VBIV5: *Module drawing for VBI V5 Tower*
- *See figure 1 for map of Japans Power Grid*
- 900-00926: *KIX5 Charger Overvoltage Issue Root Cause & Corrective Action*

Note: Japan's power grid is separated into 50Hz in Japan East and 60Hz in Japan West. The following map depicts this. This needs to be considered when designing equipment for use in Japan.



Figure 1- Japan Power Grid (Japan East 50Hz versus Japan West 60Hz)

3. Relevant Codes

The requirements detailed in this document are intended to be in compliance with all local and national electrical codes as applying to each individual building location. If local or national codes are more stringent than requirements specified herein, the supplier / contractor shall ensure compliance with the local / national legal requirement. In the absence of a relevant local/national code, the electrical contractor shall determine the appropriate standards or norms to be followed, subject always to the client's approval. The electrical contractor should also satisfy themselves as to any additional requirements as may be laid down by the client in their tender package.

In the event of a conflict between the requirements contained within this document and those laid out under local or national code (or indeed those as set out elsewhere by the client), then Amazon Robotics must be notified in writing prior to any design or installation work being undertaken.

4. Acronyms

- **AFT** – Amazon Fulfillment Technology
- **AMZL** – Amazon Logistics
- **AR** – Amazon Robotics
- **BOM** – Bill of Materials
- **DU** – Robotic Drive Unit
- **FCX** – FC Execution System
- **MSC** – Main Safety Controller

- **PSC** – Perimeter Safety Controller
- **SSD** – Sub Same Day
- **VBI** – Visual Bin Inspection

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5. General Requirements

Unless otherwise noted these requirements apply to all power and data drops.

General Requirements/Specifications	
5.1 Power Drops – Installation	1. Power drops shall be hung as pendants / flex drops. Appropriate strain reliefs shall be used so that tension is not applied to the conductors. Note: This is the preferred method for power drop installation. Refer to section 14.6
	2. Pendant cables shall use Nylon tie straps or Velcro to secure to fence for strain relief. See section 14.6 figure 7 for reference pictures of acceptable installation. If Nylon tie straps are used, they shall be trimmed in a manner that does not introduce sharp edges.
	3. Pendants shall attach to the fence with tie straps for strain relief. The power drops shall be secured to the fence using a method that does not require an electrician or the junction box to be opened in the event the drop needs to be moved. See section 14.6 figures 7 & 8 for reference pictures of acceptable installation.
	4. The power drops shall be secured to the fence using a method that does not require an electrician or the junction box to be opened in the event the drop needs to be moved.

General Requirements/Specifications	
5.2 Power Drops - Labeling	1. Labels must be sufficiently durable to withstand a 15 second mineral spirit rub, followed by a 15 second water rub with a soft cloth test ² . Labels must remain legible with no curling of edges permitted.
5.3 Power Drops – Enclosures	1. Maximum Size: 150mm x 150mm x 104mm
	2. Includes: faceplate, flat ring, and protective bushing
	3. All faceplates to be of a consistent color.
	4. All enclosures (outlet, junction boxes) must be rated a minimum of IP2x.33

² Note: An example label that meets these requirements is Brady B-595 permanent adhesive labels.

5.4 Power Drop Locations	1. For <u>Amazon Robotics Workstations and PSC's</u>, the heights of the power drops shall be located based on the following requirements based on Amazon building type. a. Sortable Fulfillment Centers i. Amazon Robotics Workstations can accommodate a receptacle location that is 600mm above the service level. It is permissible to move the receptacle to accommodate better cable management, or to move the drop closer to the station if required (horizontal or vertical). ii. PSC's can accommodate a receptacle location that is 600mm (to the bottom of the receptacle) above the service level. b. Pantry Fulfillment Centers i. Amazon Robotics Workstations can accommodate a receptacle location that is 2130mm above the service level. ii. PSC's can accommodate a receptacle location that is 600mm above the service level. c. Mobile Sortation Solution – Elevated Floor i. AR Workstations can accommodate a receptacle location that is 0mm above the service level. It is permissible to move the receptacle to accommodate better cable management. ii. PSC's can accommodate a receptacle location that is 600mm above the service level. d. Mobile Sortation Solution – Ground Floor i. AR Workstations can accommodate a receptacle location that is 0mm above the service level. It is permissible to move the receptacle to accommodate better cable management. ii. PSC's can accommodate a receptacle location that is 600mm above the service level. e. Pallet Sortation (S-Drive) i. AR Workstations can accommodate a receptacle location that is 600mm above the service level. ii. PSC's can accommodate a receptacle location that is 600mm above the service level. f. Air Gateway i. AR Workstations can accommodate a receptacle location that is 0mm above the service level. ii. PSC's can accommodate a receptacle location that is 600mm above the service level. g. Middle Mile Sortation Center i. AR Workstations can accommodate a receptacle location that is 600mm above the service level. ii. PSC's can accommodate a receptacle location that is 600mm above the service level. h. AMZL Delivery Station i. AR Workstations can accommodate a receptacle location that is 600mm above the service level. ii. PSC's can accommodate a receptacle location that is 600mm above the service level. 2. NOTE: The power drops shall be secured to the fence using a method that does not require an electrician or the junction box to be opened in the event the drop needs to be moved.
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General Requirements/Specifications	
	NOTE: The heights above are with respect to the service level (i.e. finished floor) to the bottom of the receptacle box. - For <u>Chargers</u>, the receptacles and power distribution boxes shall be located at least 1200mm and no greater than 1900mm above the service level. - For <u>VBI</u> power and data drop locations, see Section 12.
5.5 Power Drops – Voltage Drop	The voltage drop from supply to termination of a run shall not exceed 5%.
5.6 Data Drops – Location	- Refer to the system layout drawing for locations of all data drops. - For <u>Amazon Robotics Workstations and PSC's</u>, the data drops shall be located above the fence line, mounted to the ceiling, or located in a wire tray above the power and data drop survey point unless otherwise specified. - For buildings that have high bay ceilings (>3658mm), like sub-same day (SSD) buildings or similar, where it may not be feasible to install data drops in the ceilings or a wire tray above the fence line, it is acceptable to for the data drops to be installed at 2130mm above the finished floor on the fence. Note: The data drop shall use flexible conduit and be secured to the fence using a tie strap (or similar temporary fastening method) that may be easily removed unless otherwise dictated by local regulations.
5.7 Data Drops - Labels	Labels must be sufficiently durable to withstand a 15 second mineral spirit rub, followed by a 15 second water rub with a soft cloth test³. Labels must remain legible with no curling of edges permitted.
5.8 Data Drop-Enclosures	- Size: 100mm x 100mm x 54mm - Includes: faceplate, flat ring, and protective bushing - All faceplates to be of a consistent color EXCEPT the faceplate for the PROVISIONING network, which should be a contrasting, high-visibility color. - All enclosures (outlet, junction boxes) must be rated a minimum of IP2x.

³ Note: An example label that meets these requirements is Brady B-595 permanent adhesive labels.

6. Amazon Robotics Workstations

Workstations are the primary interface for executing work within the AR system. They provide access for getting product into and out of the AR System. All stations consist of a power and data drop for connecting a station PC and associated peripherals. There is one PSC to service one or more workstations. Refer to section 7 for power/data drops for the PSC. The location of the PSC is called out separately on the layout. **The requirements below are per workstation.**

Amazon Robotics Workstation Specifications	
6.1 Power	1. One 100VAC, 50 or 60Hz ⁴ , 15 Amp circuit with duplex (2) JIS C 8303 Class I receptacles powering the Amazon Robotics workstation peripherals. Maximum current draw is 8.2A. When ganging multiple outlets together, it shall be done in accordance with the local / regional electrical codes.
	2. One 100VAC, 50 or 60Hz ⁵ , 15 Amp circuit with duplex (2) JIS C 8303 Class I receptacles (see section 14.3 for Electrical Convenience Outlets). Convenience receptacles may use the same circuit as the 100 VAC, 15 Amp station power circuit as long as the Amazon Robotics Workstation current draw requirement (8.2A) is considered, but must NOT be shared with PSC circuits (See section 7). When ganging multiple outlets together, it shall be done in accordance with the local / regional electrical code.
	3. Note: Recommended implementation of 6.1 items 1 & 2 above are to use one double-duplex (quad) JIS C 8303 Class I receptacles per station.

⁴ 50Hz applies to Japan East and 60Hz applies to Japan West.

⁵ 50Hz applies to Japan East and 60Hz applies to Japan West.

Amazon Robotics Workstation Specifications	
6.2 Data	Two CAT6 data cable connections are required for each AR workstation; these are described below in detail: • One CAT6 data cable connection to IDF is needed for the 'STATION' network. • One CAT6 data cable connection to the IDF is needed for the 'STATION' network spare. 'STATION' network & 'STATION' network spare data cable connections shall terminate in the same data device at the Amazon Robotics Workstation location. 'STATION' network & 'STATION' network spare data cable connections can terminate on the same patch panel in adjacent data port positions at the IDF. Amazon Robotics Workstation 'STATION' network data cable connections shall not terminate on the same patch panel(s) as PSC 'SAFETY' network data cable connections at the IDF.
6.3 Location	Refer to workstation layout drawings for placement. These vary per station type.

7. PSCs

Perimeter Safety Controllers (PSCs or SFBs) are installed near gate locations and AR Workstations. There may be one PSC connected to multiple controls at nearby stations and gates. These controls include E-stop, light stack and local resets. The floor layout drawing calls out locations for each PSC (or SFB). **The requirements below are per PSC.**

Specifications	
7.1 Power	One 100VAC, 50 or 60 Hz⁶, 20 Amp circuit with a single NEMA L5-20R receptacle is required. Maximum current draw is approximately 1.25 Amps. When ganging multiple outlets together, it shall be done in accordance with the local / regional electrical codes. This circuit, including power and neutral, shall only be shared only with other safety equipment receptacles (PSCs). This circuit shall not be shared with general building power, VBI or any Amazon Robotics Work Station power.
7.2 Data	One CAT6 data cable connection to the IDF is needed for the 'SAFETY' network. Amazon PSC 'SAFETY' network data cable connections shall not terminate on the same patch panel(s) as Amazon Workstation 'STATION' network or Robin Station 'STATION' & 'SAFETY' network data cable connections at the IDF.

8. Battery Charging Stations

Charging stations provide high power for battery recharging of the AR drive units and are typically situated in clusters of four around a building column. Below are 2 implementation methods to support power drops for battery charging stations; branch circuit and power distribution methods. Typically implementation is the power distribution method especially to support charger clusters. The branch circuit method is an option in buildings where only 1 charger may be required.

NOTE: The Battery charging stations require a 400VAC input, which is not a standard 3-phase supply voltage in the JP region. A building-wide 200V-to-400V step-up transformer must be provided to supply the 400V service needed for the battery charging stations.

NOTE: It is possible for all chargers to pull the max current simultaneously.

NOTE: Due to issues discovered at KIX5 related to overvoltage issues, a white paper was written that provides root cause and corrective action for this and provides guidance for properly sizing transformers for the charger 3-phase alternating current (AC) power. See section 2.0 for white paper document number.

⁶ 50Hz applies to Japan East and 60Hz applies to Japan West

Battery Charging Stations Specifications	
8.1 Branch Circuit Method	1. One 50 or 60 Hz ⁷ 400VAC, 3-phase, 16 Amp circuit with fused disconnect feeding one IEC 60309 3P+E 6h Red 16Amp receptacle 600mm above the finished floor PER CHARGER . Maximum current draw is approximately 13.5 Amps. The circuit must have a 3-pole Type D breaker with not higher than 16 amp current rating.
	2. One 100VAC, 50 or 60Hz ⁷ , 15 Amp, circuit with one or two outlet receptacles for convenience use PER CHARGER CLUSTER (JIS C 8303 Class I earthed) - (see section 14.3 for Electrical Convenience Outlets).. The circuit for this outlet <u>must not be shared</u> with the PSC circuits (See section 7).
8.2 Power Distribution Box Method	1. One 50 or 60 Hz ⁷ 400VAC, 3-phase, 64 or 63 Amp circuit is to be supplied to a column-mounted termination box and power distribution board. Note: If a column is not available, unistrut shall be used and configured such that it provides adequate support for the power distribution box and shall be anchored at floor level with a 4 anchor baseplate. The distribution board should be positioned at a height of 1200mm above service level, to underside (unless otherwise required by local or national code requirements). Maximum current draw is approximately 13.5 Amps PER CHARGER and one distribution panel should serve a maximum of FOUR CHARGERS .
8.3 Power Distribution Box Specifications	1. Provided by electrical contractor
	2. Enclosure is at least IP2x or NEMA 1 rated
	3. Maximum enclosure dimensions are 0.61m tall, 0.51m wide, and 0.25m thick; See Appendix 14.2 for an example
	4. Back panel for enclosure
	5. Input is 50 or 60Hz ⁷ , 400VAC, 3-phase power
	6. 63 or 64 Amp, 3-pole main circuit breaker on the input
	7. Four 16 Amp, 3-pole Type D breakers
	8. Four externally-accessible, lockable, power disconnect switches on the side of the enclosure; one for each 16 Amp 400VAC branch circuit, labeled externally to identify the matching receptacle.
	9. Four externally-accessible, IEC 60309 3P+E 6h Red 16Amp, 3-phase power receptacles on the bottom of the enclosure, labeled externally to identify the matching disconnect switch.
	10. 2KVA or greater 400/100VAC power transformer with protective fusing
	11. One or two externally-accessible, 100VAC, 50 or 60Hz ⁷ , 15Amp, JIS C 8303 Class I earthed convenience outlet on the side of the enclosure.
	12. Grounding lug

⁷ 50Hz applies to Japan East and 60Hz applies to Japan West

Battery Charging Stations Specifications	
	13. Custom engraved legend plates as required.
	14. Terminal blocks, wire, and wire duct as needed.
	15. Compliance with all applicable regulatory standards
8.4 Box Locations	Location of the distribution box (column face or unistrut and horizontal location) will be called out on the AR layout and shall be located on the same side of the fence line as the location of the charger. Please consult the layout before final installation to avoid interference with the AR active floor. The feed termination junction box and Power Distribution Unit should be mounted on the same side of the column when possible. An example is provided in Appendix Error! Reference source not found..

9. FCX

The FCX cabinet has the server and network equipment required to host the DU movement control software of the AR System and resides in the MDF room. **Refer to network plan for the number of FCX cabinets per site.**

FCX Specifications	
9.1 Power	1. One 200VAC, 30 Amp circuit with an approved disconnect box in proximity to a single 200V L6-30 receptacle for primary ATS (APC AP7732 or equivalent). This circuit must be connected to the MDF UPS and may be shared only with other UPS outlets, not general building power. Maximum current draw is approximately 24 Amps. 2. One 200VAC, 30 Amp circuit with an approved disconnect box in proximity to a single L6-30 receptacle for primary ATS (APC AP7732 or equivalent). This circuit is connected to general building power. Maximum current draw is approximately 24 Amps. 3. One 200VAC, 30 Amp circuit with an approved disconnect box in proximity to a single 200V L6-30 receptacle for secondary ATS (APC AP7732 or equivalent). This circuit must be connected to the MDF UPS and may be shared only with other UPS outlets, not general building power. Maximum current draw is approximately 24 Amps. 4. One 200VAC, 30 Amp circuit with an approved disconnect box in proximity to a single L6-30 receptacle for secondary ATS (APC AP7732 or equivalent). This circuit is connected to general building power. Maximum current draw is approximately 24 Amps. 5. Cabinet ground required (PE) – it is acceptable to tie into the MDF ground loop. 6. UPS and generator must be capable of providing a minimum of 1hr of uptime in the event of a power outage.
9.2 Data	1. Refer to network plan. Data connections are routed within the MDF.
9.3 Environment	1. Room temperature: 10 – 32.22 degrees °C 2. Humidity: 8 – 80%
9.4 Location	1. Minimum access clearance of 920mm behind and 920 mm in front of FCX cabinet.

10. MSC

The MSC is the central controller for all safety equipment on a given AR floor and communicates status to the MHS. Depending on configuration, one MSC is required per one or two FC Zones. In addition, a warm standby MSC per building may be specified. All are located in the MDF room and are shown in the IT Playbook layout plan for the MDF. **The requirements below are per MSC.**

MSC Specifications	
10.1 Power	One 100 VAC, 50 or 60Hz ⁸ , 20 Amps, NEMA L5-20R receptacle. This circuit should originate from the UPS backup within the MDF, and be connected to the EPO. Maximum current draw is approximately 2 Amps.
10.2 Data	Refer to the network plan. Use CAT6 data cables for all connections. Ensure sufficient service loop is available for swapping to a standby MSC for failover, or use a patch panel.
10.3 Environment	1. Heat displacement: ~825 BTU/hr. per MSC.
	2. Room temperature: 10 – 32.22 degrees °C
	3. Humidity: 8 – 80%
10.4 Location	1. Refer to the MSC Dimensions in the Appendix for clearance and elevation requirements
	2. Must be mounted vertically according to the assembly instructions.
	3. Minimum access clearance 920mm in front of MSC panel.
	4. Power cord provided is 3048mm long, located at top right of cabinet. See Appendix

⁸ 50Hz applies to Japan East and 60Hz applies to Japan West

11. Fire Alarm

Fire Alarm Specifications	
11.1 Location	The MSC and a Fire Alarm Relay are recommended to be co-located in the same room and no further than 1524mm apart from each other to allow integration with a 20m cable included with the MSC assembly.
11.2 Required Space	Wall mounted
11.3 Theory of Operation	Two +24VDC signals are required as inputs to the MSC to satisfy a normal operation state (no fault condition). When a fault occurs, both +24VDC signals shall be removed as inputs to the MSC to satisfy a fault state. To reset this fault, both +24VDC signals shall be engaged within 0.5 seconds of each other, otherwise, the fault condition will persist.
11.4 Connection to Fire Alarm	1. Use a relay with two dry contacts that will satisfy 11.3
	2. Current sinking input type
	3. Voltage, on-state input = 11VDC – 30VDC
	4. Voltage, off-state input, max = 5VDC
	5. Current, on-state input, min = 3.3mA
	6. Current, off-state, max = 1.3mA
	7. Both relay contacts are closed during normal operation; no fault condition
	8. Both relay contacts are open in the event of a fire alarm; fault condition
	9. See the Appendix for Fire Alarm Electrical Connections

12. VBI Tower

This section provide powers and data wiring requirements for the VBI tower. The AFT team will not provide or install any of the materials indicated by this document unless otherwise noted. The requirements pertain to the following:

VBI Tower Specifications	
12.1 General	Please reference local site real estate electrical requirements for infrastructure, containment, labelling and voltage drop threshold guidelines. Contact local site facilities manager for these.
12.2 Power⁹	One 100VAC, 50 or 60Hz ¹⁰ , 15 Amp circuit with a single receptacle (JIS C 8303 receptacle) for the VBI tower is required. Maximum current draw is 7.6A.
12.3 Data	One CAT6 data cable connection to IDF is needed for the VBI Tower.
12.4 Location¹¹	1. Electrical Outlets: VBI towers can accommodate a receptacle location that is 2130mm above the service level.
	2. Data Port: ports being joist/ceiling mounted, recommended implementation is to the lowest ceiling truss when referring to joist/ceiling mounted,(for mezzanine structures) and no greater than 2130mm hanging pendants for non-mezzanine locations accessible from outside the fence.

⁹ Refer to local/regional electrical standards for line voltage; also install protective earthing conductor by means of the VBI power cord connected to a socket-outlet with earthing connection.

¹⁰ 50Hz applies to Japan East and 60Hz applies to Japan West

¹¹ Refer 1737-SUPP-VBIV5 Module drawing for specific location with respect to tower.

13. **Robin**

This section provide powers and data wiring requirements for Robin.

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Robin Station Power & Data Specifications	
13.1 Power	1. The Robin station requires two separate power drops. One power drop is for station peripherals and convenience use. The other power drop is for main system power. See below for requirements for each drop.
13.1.1 Station Peripheral & Convenience Power Drop	1. One 100VAC, 50 or 60Hz ¹² , 15 Amp circuit with duplex (2) JIS C 8303 Class I receptacles for powering the Robin Workstation Peripherals. Maximum current draw is 8.2A. When ganging multiple outlets together, it shall be done in accordance with the local /regional electrical codes.
	2. One 100VAC, 50 or 60Hz ¹³ , 15 Amp circuit with duplex (2) JIS C 8303 Class I receptacles for convenience use. Convenience receptacles may use the same circuit as the 100 VAC, 15 Amp station power circuit as long as the Robin station current draw requirement (8.2A) is considered, but must NOT be shared with PSC circuits (See section 7). When ganging multiple outlets together, it shall be done in accordance with the local / regional electrical code.
	3. Note: Recommended implementation of 13.1.1 items 1 & 2 above are to use one double-duplex (quad) JIS C 8303 Class receptacles per station.
13.1.2 Station Peripheral & Convenience Power Drop Location	1. The station peripheral & convenience power drop shall be located at least 820mm above the service level above the single-phase survey point and shall be installed in accordance with local / regional electrical code.
13.1.3 Main System Power Drop	1. One 400VAC 3phase 50Hz 20 Amp circuit in proximity to a single IEC 60309 3P+N+E Red 32 Amp receptacle (5 terminal outlet). Maximum current draw is 15 Amps PER STATION. The circuit must have a 3-pole type D breaker with not higher than 20A current rating.
13.1.4 Main System Power Drop Location	1. The station main system power drop shall be located at least 820mm above the service level above the 3-phase survey point and shall be installed in accordance with local / regional electrical code.

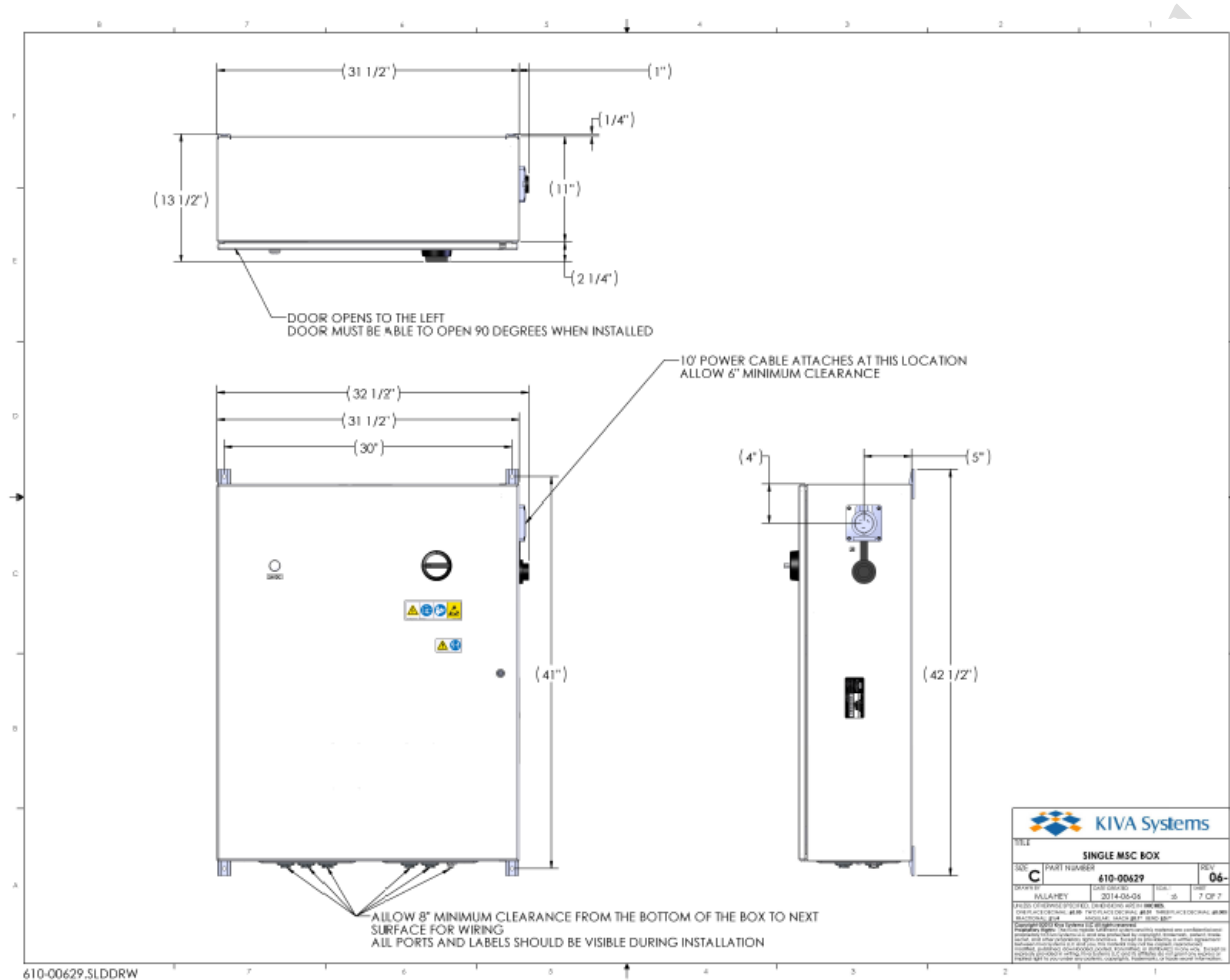
¹² 50Hz applies to Japan East and 60Hz applies to Japan West.¹³ 50Hz applies to Japan East and 60Hz applies to Japan West.

13.2 Data	Data Port: For <u>Robin Stations</u>, the data drops shall be located above the fence line, mounted to the ceiling, or located in a wire tray above the power and data drop survey point. Two CAT6 data cable connections are required for each AR workstation; these are described below in detail: • One CAT6 data cable connection to IDF is needed for the ‘STATION’ network. • One CAT6 data cable connection to the IDF is needed for the ‘SAFETY’ network. Robin Station ‘STATION’ & ‘SAFETY’ network data cable connections shall terminate in the same data device at the Robin Station location. The Robin Station ‘SAFETY’ network data cable connection is a separate network connection than a PSC ‘SAFETY’ network data cable connection. Robin Station ‘STATION’ & ‘SAFETY’ network data cable connections can terminate on the same patch panel in adjacent data port positions at the IDF.
13.2.1 Data Drop Location	Data Port: For <u>Robin Stations</u>, the data drops shall be located at least 820mm above the service level above the power and data drop survey point.

14. APPENDIX

14.1 MSC Dimensions

Refer to the contract BOM for assembly instructions. The following image (from 610-00629) is for reference only and is not a substitute for the controlled document.



FOR REFERENCE ONLY

14.2 Battery Charging Station Power Distribution Box

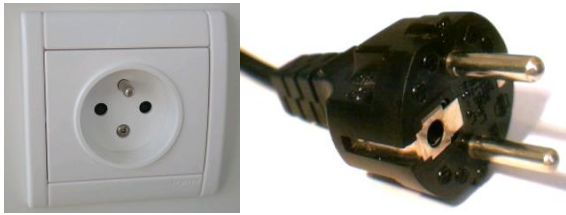


Figure 2- Power Distribution Box Example (For Reference Only)

14.3 Electrical Convenience Outlets

Definition of electrical outlet is specific to the country of application. Please refer to the below for the specific country and electrical outlet details:

Country	Electrical Outlet Type	Electrical Plug Type
Austria	Schuko CEE7/4	Schuko CEE7/7
Belgium	Schuko CEE7/5	Schuko CEE7/7
Bulgaria	Schuko CEE7/4	Schuko CEE7/7
Cyprus	BS 1363 or Schuko CEE7/4	BS 1363 or Schuko CEE7/7
Czech Republic	Schuko CEE7/5	Schuko CEE7/7
Denmark	Schuko CEE7/5	Schuko CEE7/7
Estonia	Schuko CEE7/4	Schuko CEE7/7
Finland	Schuko CEE7/4	Schuko CEE7/7
France	Schuko CEE7/5	Schuko CEE7/7
Germany	Schuko CEE7/4	Schuko CEE7/7
Greece	Schuko CEE7/4	Schuko CEE7/7
Hungary	Schuko CEE7/4	Schuko CEE7/7
Ireland	BS 1363	BS 1363
Italy	Schuko CEE7/4 or CEI 23-16/VII	Schuko CEE7/7 or CEI 23-16/VII
Japan	JIS C 8303 Class I earthed	JIS C 8303 Class I earthed
Latvia	Schuko CEE7/4	Schuko CEE7/7
Lithuania	Schuko CEE7/4	Schuko CEE7/7
Luxembourg	Schuko CEE7/4	Schuko CEE7/7
Malta	BS 1363	BS 1363
Netherlands	Schuko CEE7/4	Schuko CEE7/7
Poland	Schuko CEE7/5	Schuko CEE7/7
Portugal	Schuko CEE7/4	Schuko CEE7/7
Romania	Schuko CEE7/4	Schuko CEE7/7
Slovakia	Schuko CEE7/5	Schuko CEE7/7
Slovenia	Schuko CEE7/4	Schuko CEE7/7
Spain	Schuko CEE7/4	Schuko CEE7/7
Sweden	Schuko CEE7/4	Schuko CEE7/7
United Kingdom	BS 1363	BS 1363

Schuko CEE 7/4 (compatible with CEE 7/7 Plug)**Schuko CEE 7/5 (compatible with CEE 7/7 Plug)****BS 1363****NEMA 5-15 / JIS C 8303 Class I earthed****IEC 60309 6H BLUE 2P+E**

IEC 60309 6H RED 3P+E**FOR REFERENCE ONLY****14.4 Facility Alarm Electrical Connections**

Representative of the Fire Alarm Electrical Connection in normal state (no fault condition)

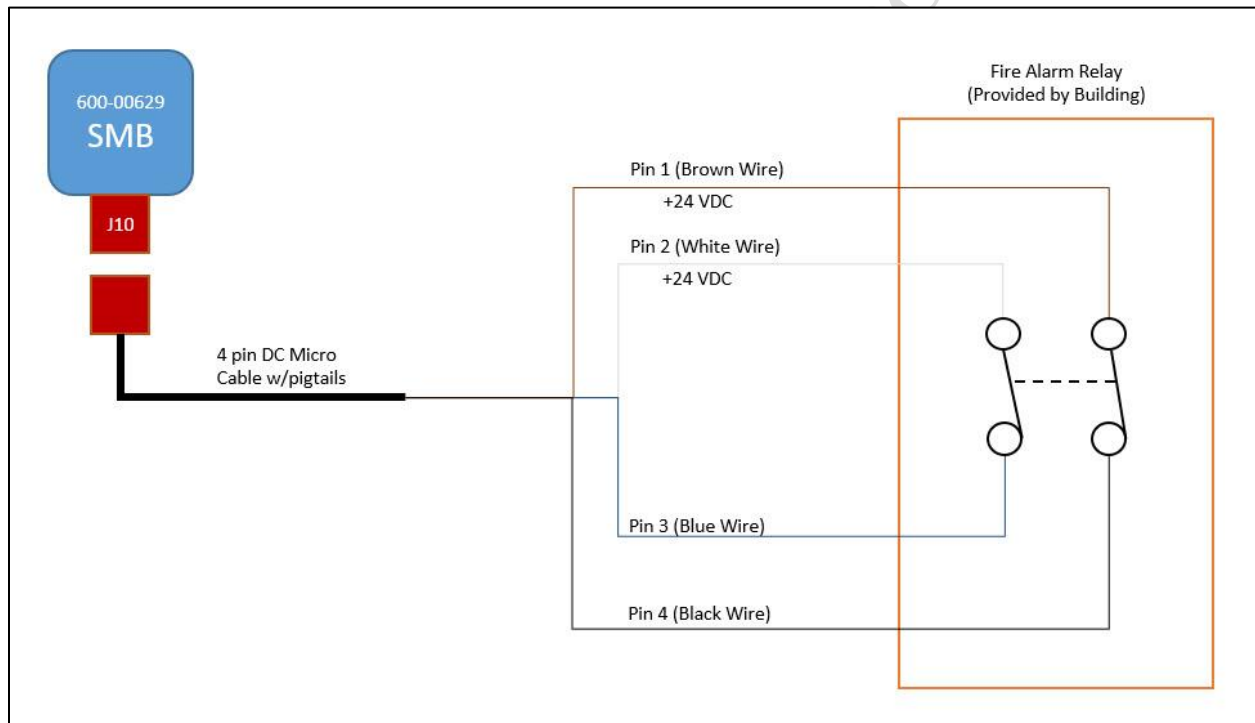


Figure 3- Fire Alarm Connection

14.5 Data Drops in the Ceiling

The following figures are intended to show examples of power drops in the ceiling for clarification purposes.



Figure 4- Data Drop in Ceiling - Example 1



Figure 5- Data Drop in Ceiling – Example 2

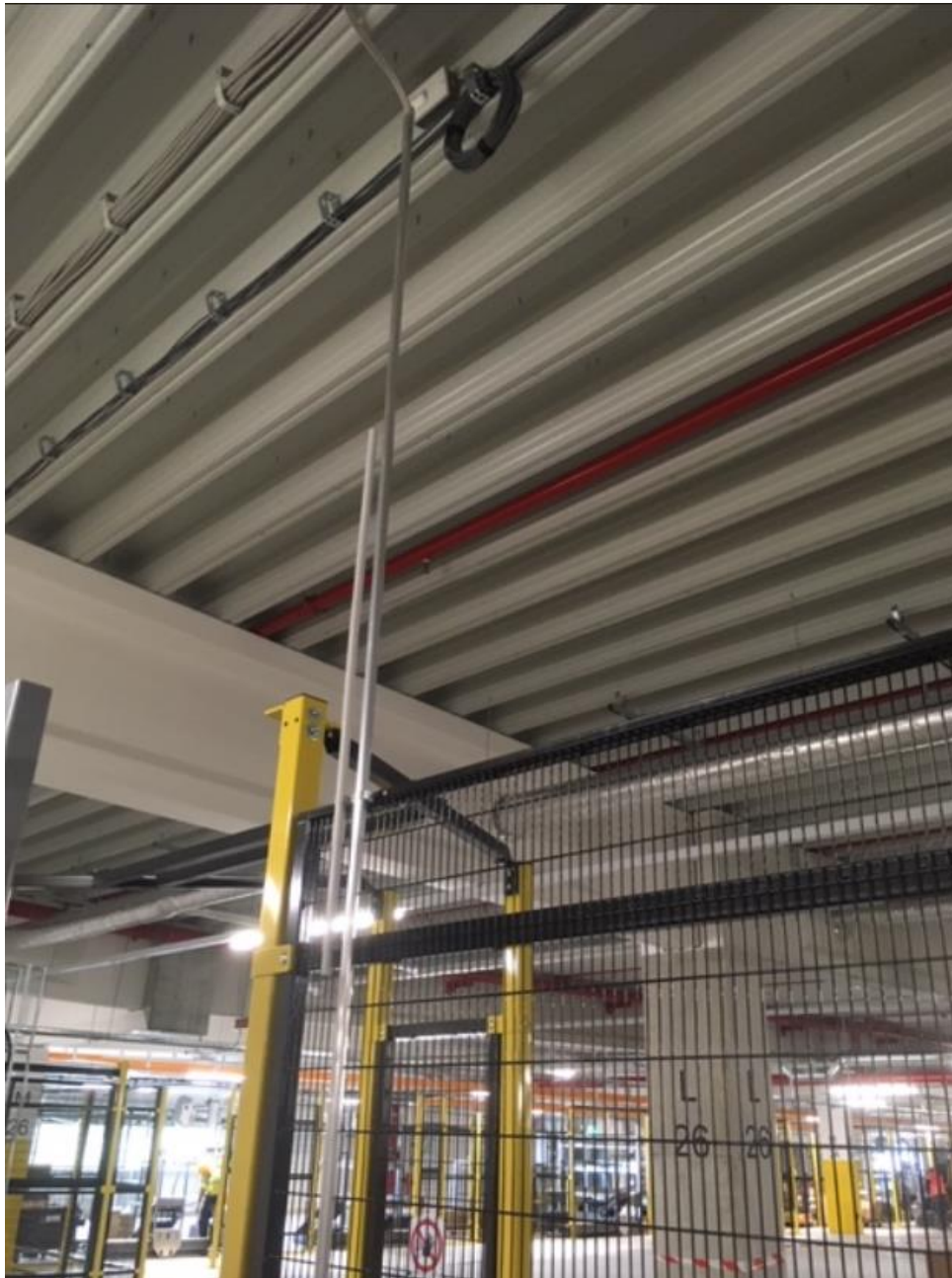


Figure 6- Data Drop in Ceiling – Example 3

14.6 Example Flex Drop Installations

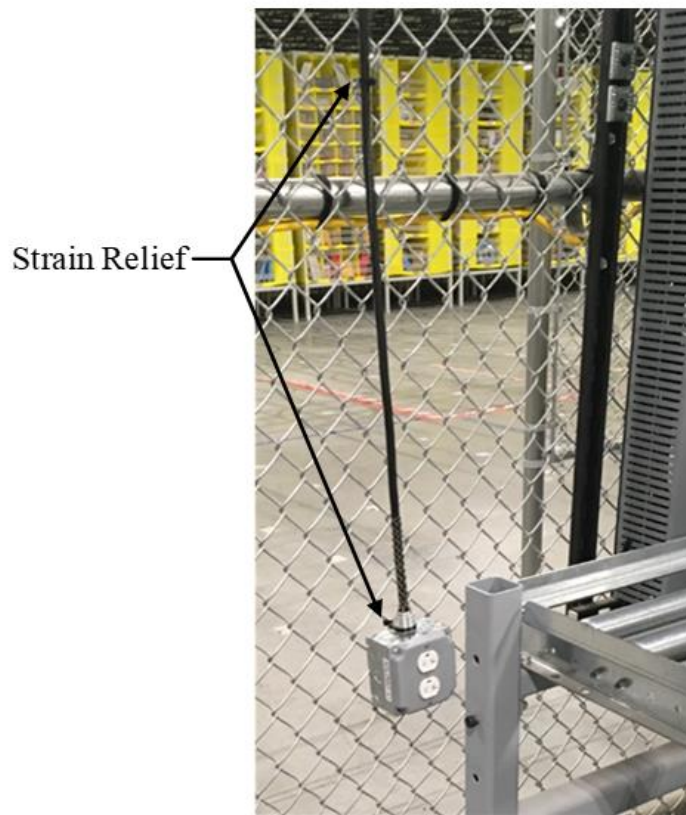


Figure 7- Flex Drop with Tie Strap Strain Relief